1610 ▸ 1630

Calculating machines
See pages 124–125

Io

Europa

Ganymede

Callisto

1610

Moons of Jupiter

Observing the night skies through the telescope he had built himself, Galileo Galilei (see pp.68–69) noticed three small stars near the planet Jupiter that changed position over a period of time. He realized they were moons, or satellites (objects that orbit a planet or star), circulating the planet, and later identified a fourth. These four were the brightest of Jupiter's moons, often called the Galilean moons. Galileo's observation contradicted the Church's teaching that everything in the Universe rotated around Earth.

1614

Multiplying numbers

Scottish mathematician John Napier introduced logarithms, which are used to multiply and divide very large numbers—especially useful in astronomy. In 1617, he introduced another aid to calculation—a set of rods divided into sections and marked with digits, which became known as Napier's bones.

Set of movable rods inscribed with digits for multiplying and dividing

Fixed column divided into numbers from 1 to 9

Napier's bones

1610 · 1615 · 1620

1620

First submarine

Cornelis Drebbel, a Dutch inventor living in England, built a submarine made of a wooden frame covered with leather. It was powered by oars. The submarine stayed underwater for three hours when given a trial run in the Thames River, but it is unclear how the oarsmen sitting inside were able to breathe.

Slide rule

Another handy aid to calculation was the slide rule, invented by English mathematician William Oughtred in 1622. Slide rules remained in use until the late 20th century, when they were replaced by pocket calculators.

Replica of Drebbel's submarine